

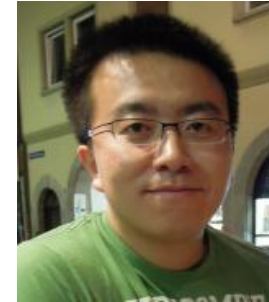
Quantum Fisher Information as a Thermal and Dynamical Probe in Frustrated spin system

Cooperators



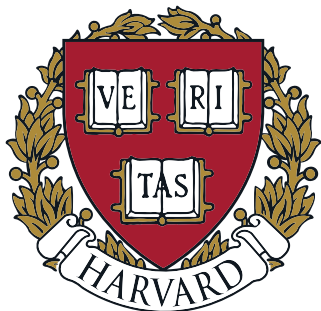
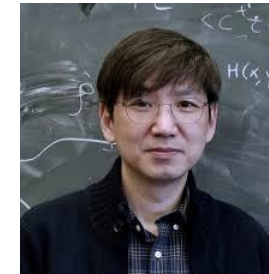
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The k -producibility state $|\psi\rangle = \bigotimes_{i=0}^M |\psi_i\rangle$
 Entanglement depth m means $m \geq k_i, \forall i$



$$|\psi\rangle = (|\uparrow\rangle_0 + |\downarrow\rangle_0) \otimes (|\uparrow\rangle_1 + |\downarrow\rangle_1) \otimes (|\uparrow\rangle_1 + |\downarrow\rangle_1) \otimes \dots$$



$$|\psi\rangle = (|\uparrow\downarrow\rangle_{01} - |\downarrow\uparrow\rangle_{01}) \otimes (|\uparrow\downarrow\rangle_{23} - |\downarrow\uparrow\rangle_{23}) \otimes \dots$$



$$|\psi\rangle = (|\uparrow\uparrow\uparrow\uparrow \dots\rangle + |\downarrow\downarrow\downarrow\downarrow \dots\rangle)$$



$$|\psi\rangle = (|\uparrow\downarrow\rangle_{01} - |\downarrow\uparrow\rangle_{01}) \otimes (|\uparrow\uparrow\uparrow\uparrow \dots\rangle + |\downarrow\downarrow\downarrow\downarrow \dots\rangle)$$

Quantum Fisher information

$$F_Q[\rho; \hat{O}(\mathbf{q})] = 2 \sum_{i,j} \frac{(p_i - p_j)^2}{p_i + p_j} |\langle i | \hat{O}(\mathbf{q}) | j \rangle|^2,$$

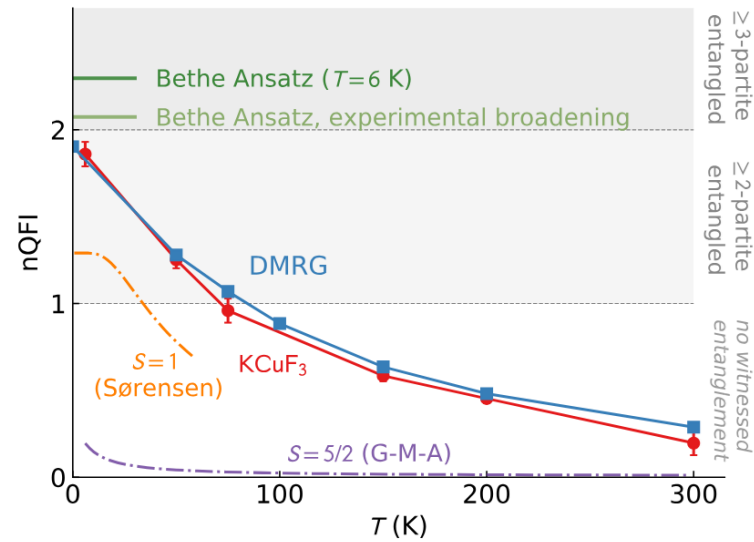
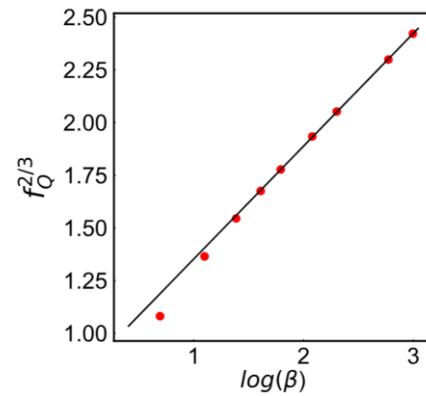
with $\rho = \sum_i p_i |i\rangle\langle i|$ and $p_i = \exp(-E_i/T)/Z$.

And

$$f_{QFI} = F_Q/N < m (\hat{O}_{max} - \hat{O}_{min})^2$$

Also,

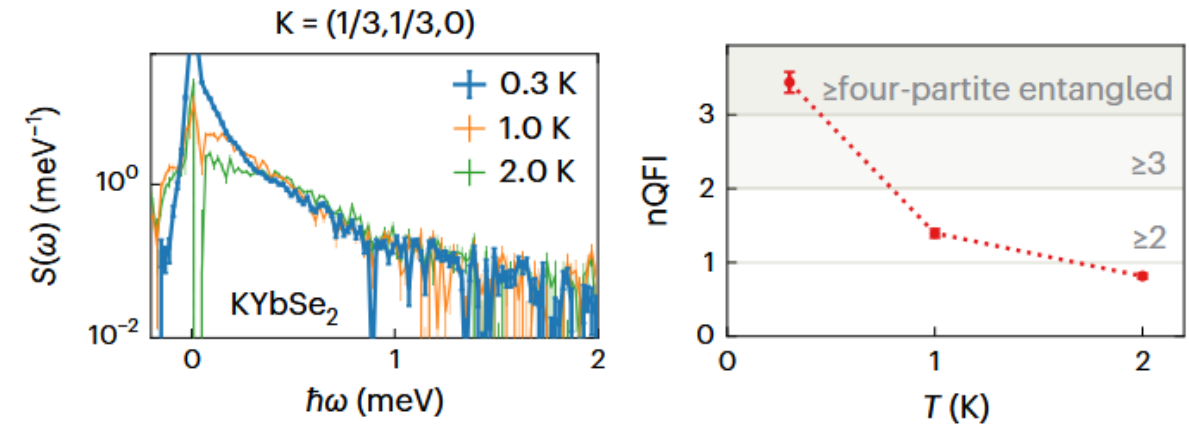
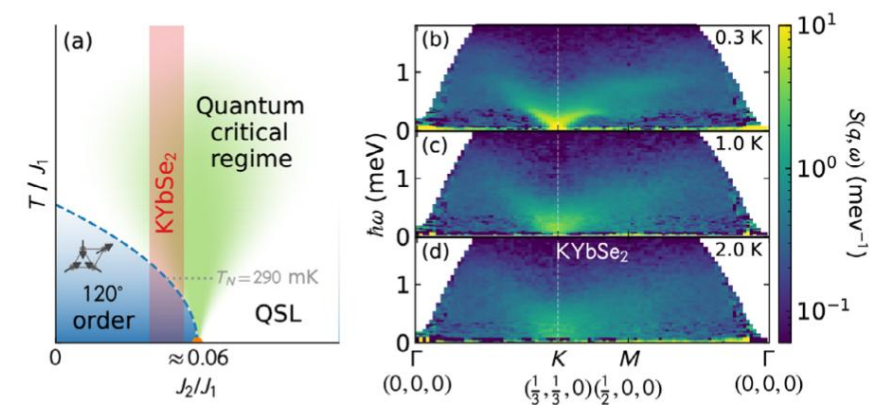
$$f_{QFI} = \int_0^\infty d\omega \, 4 \tanh\left(\frac{\omega\beta}{2}\right) \pi [1 - \exp(-\beta\omega)] A_{\hat{O}}(\beta, \omega)$$



Heisenberg spin chain

V. Menon, et al., [Phys. Rev. B 107, 054422 \(2023\)](#)

A. Scheie, et al., [Phys. Rev. B 103, 224434 \(2021\)](#)

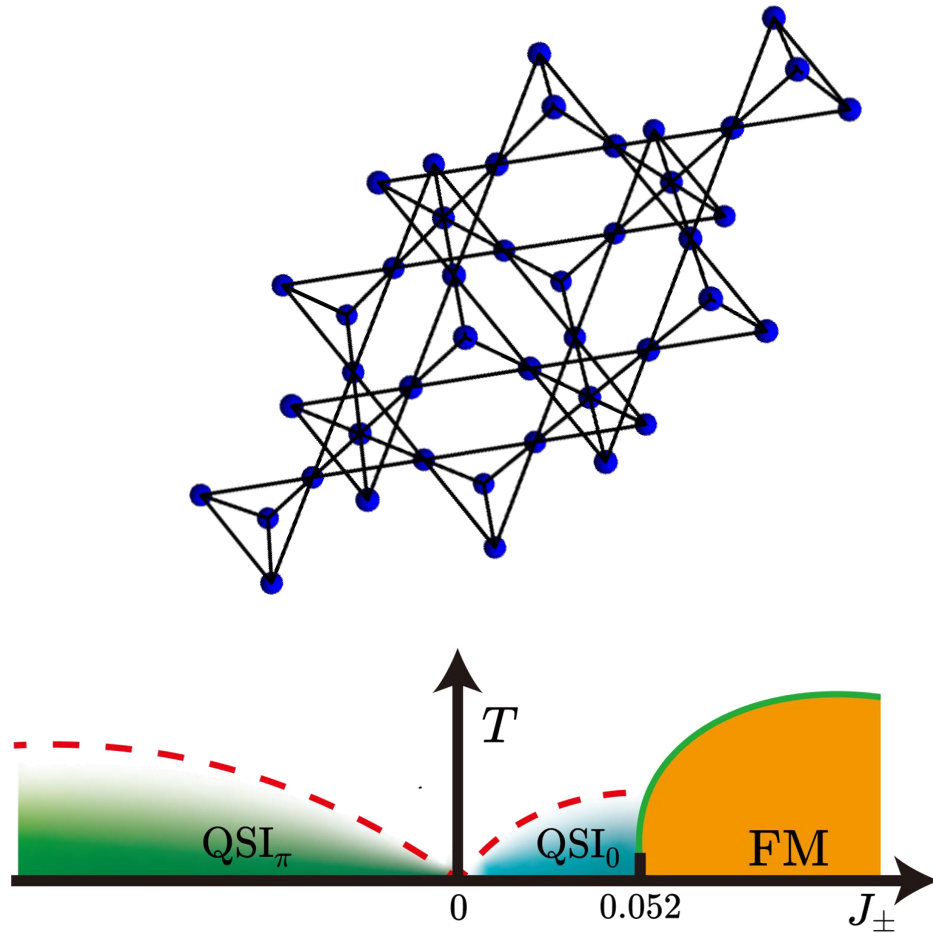


Triangle Lattice

A. O. Scheie, et al., [Nat. Phys. 20, 74–81 \(2024\)](#)

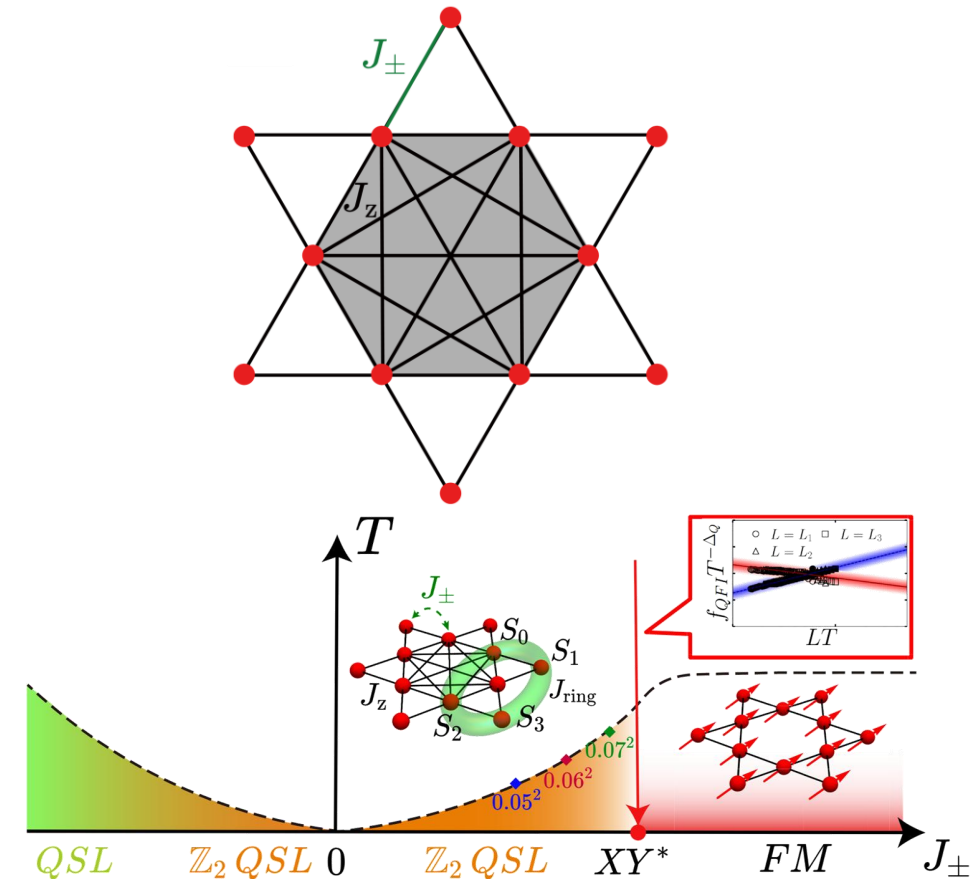
1, Pyrochlore lattice

Nat Commun 17, 7975 (2026)



2, Balents-Fisher-Girvin (BFG) model

arXiv:2603.19951 (2026)



$$H = J_z \sum_{\langle i,j \rangle} S_i^z S_j^z - J_{\pm} \sum_{\langle i,j \rangle} (S_i^+ S_j^- + S_i^- S_j^+)$$

Ring Exchange in Quantum Spin Ice regime:

$$H_{ring} = J_{ring} \sum_{\langle i,j,k,l,m,n \rangle} (S_i^+ S_j^- S_k^+ S_l^- S_m^+ S_n^- + S_i^- S_j^+ S_k^- S_l^+ S_m^- S_n^+)$$

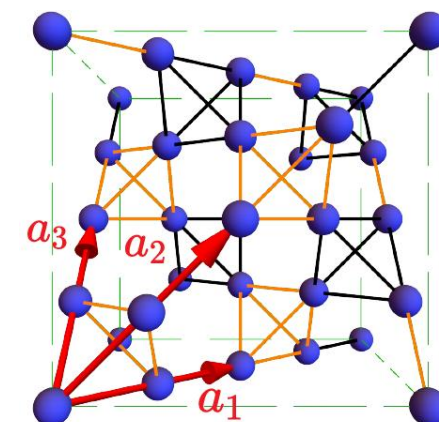
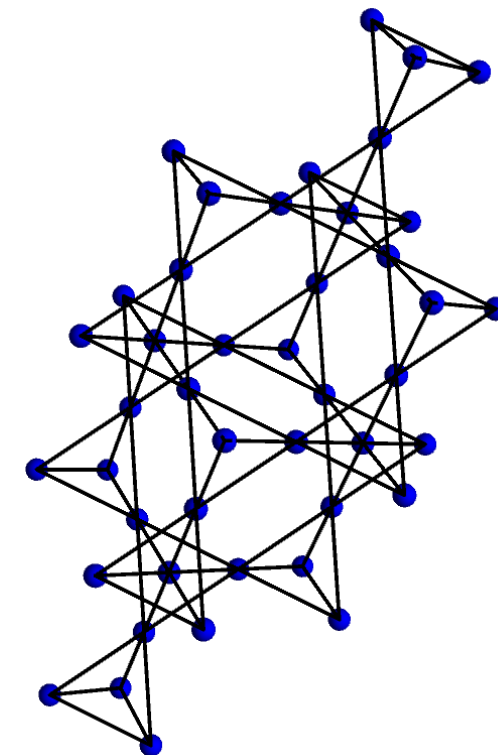
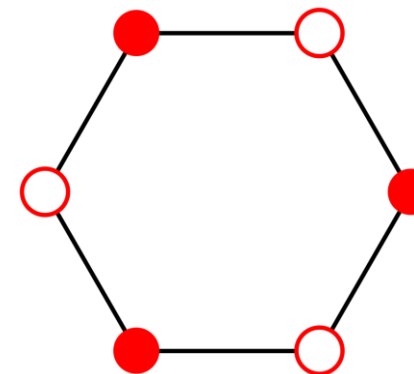
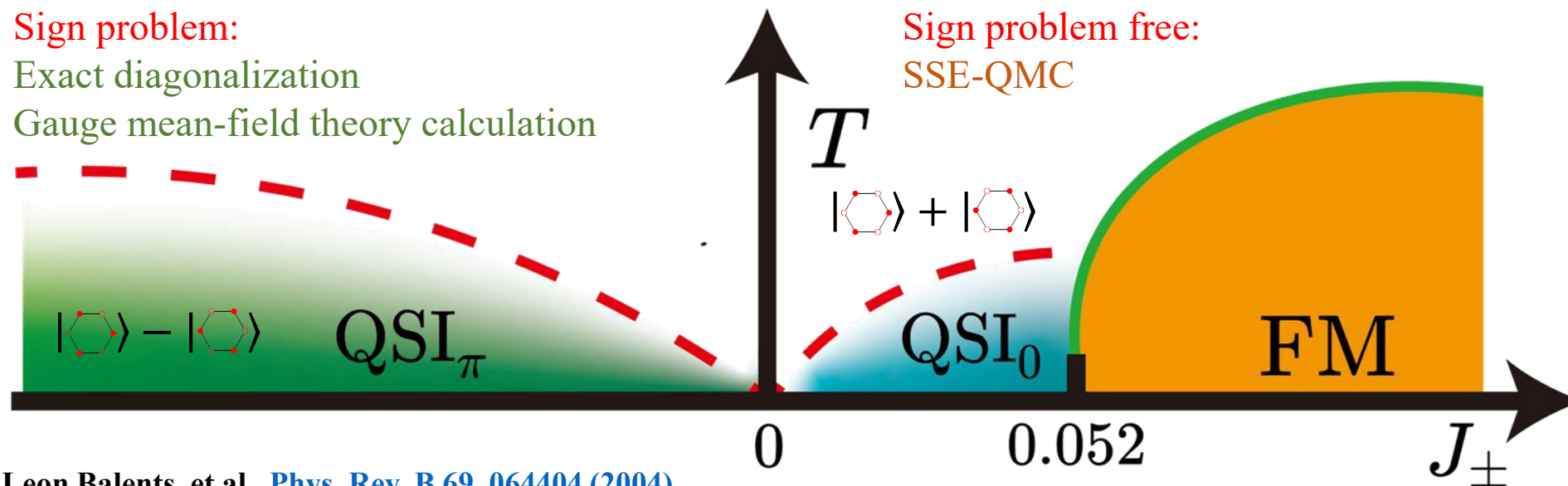
$$J_{ring} = \frac{12J_{\pm}^3}{J_z^2}, T \sim J_{ring}$$

If $J_z = 1, J_{\pm} = 0.045$, then $T \sim 0.0010935$

Sign problem:

Exact diagonalization

Gauge mean-field theory calculation



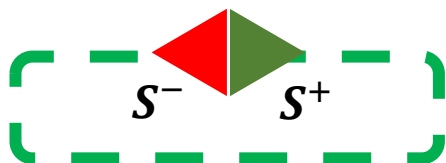
Leon Balents, et al., [Phys. Rev. B 69, 064404 \(2004\)](#)

Johannes Reuther, et al., [Phys. Rev. B .111, 134442 \(2025\)](#)

$$Z = \sum_{n=0}^{\infty} \frac{\beta^n}{n!} \sum_{\{S_n\}} \sum_{\{\alpha\}} \langle \alpha | \Pi(-H_b) | \alpha \rangle$$

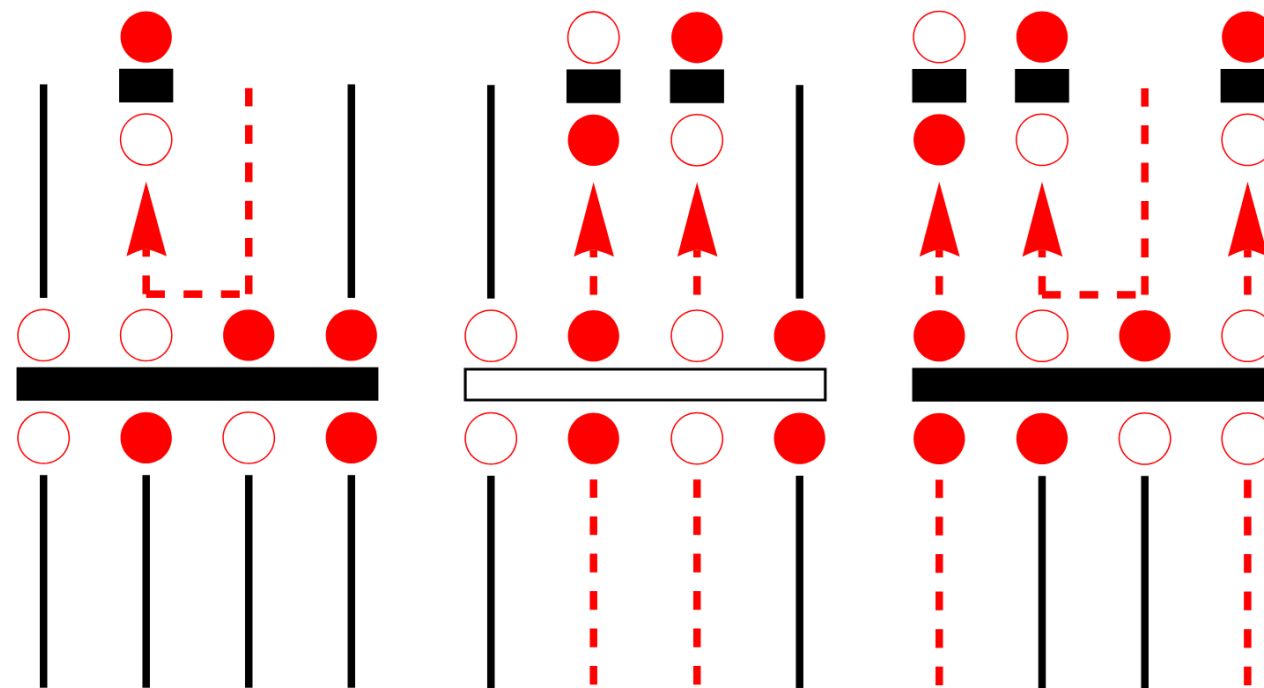
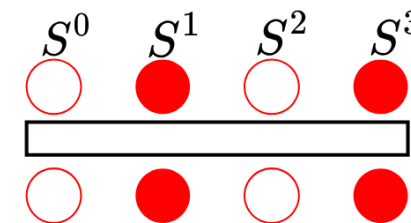
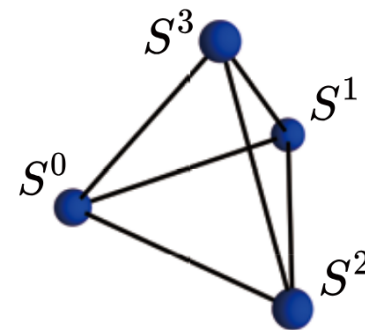
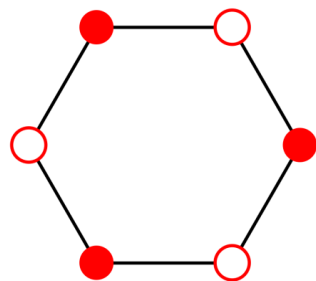
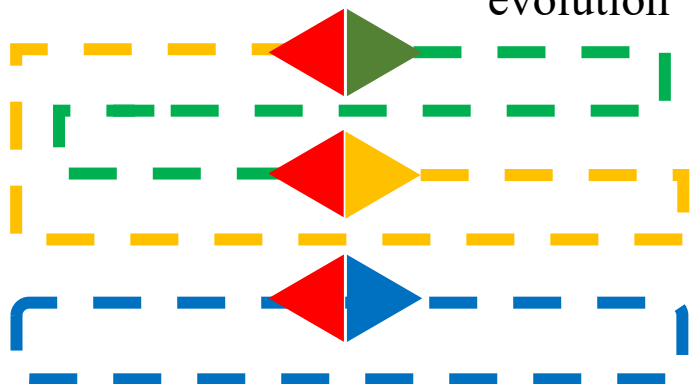
Directed loop (DL) update:

$$C_1 \xrightarrow[1 \text{ pair of } S^+ S^-]{\text{evolution}} C_2$$

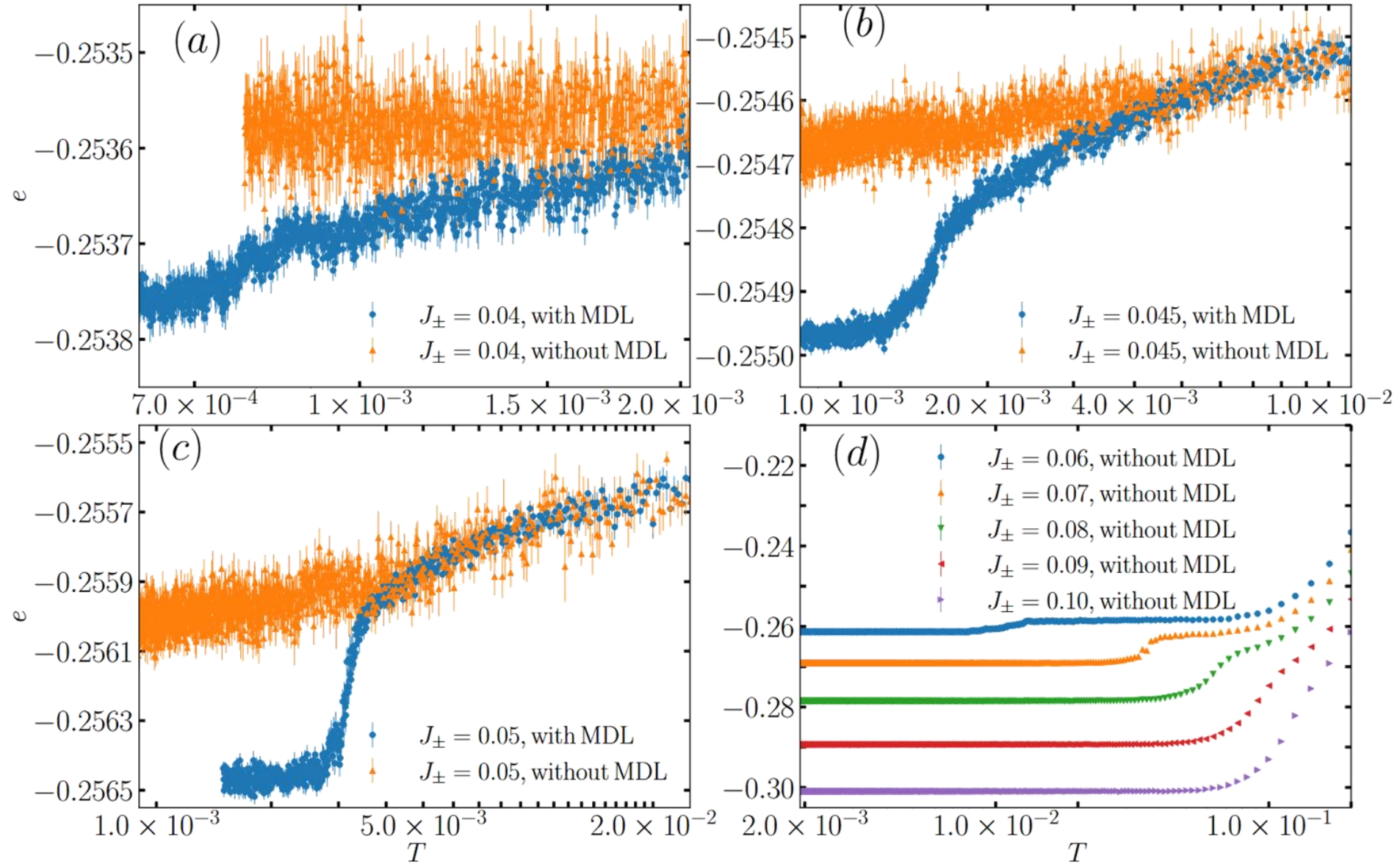


Multi-directed loop (MDL) update:

$$C_1 \xrightarrow[3 \text{ pairs of } S^+ S^-]{\text{evolution}} C_3$$



Thermal annealing process: $\beta = 1/T$ from 1 to 1500 with $\delta\beta = 1$



Exact diagonalization:

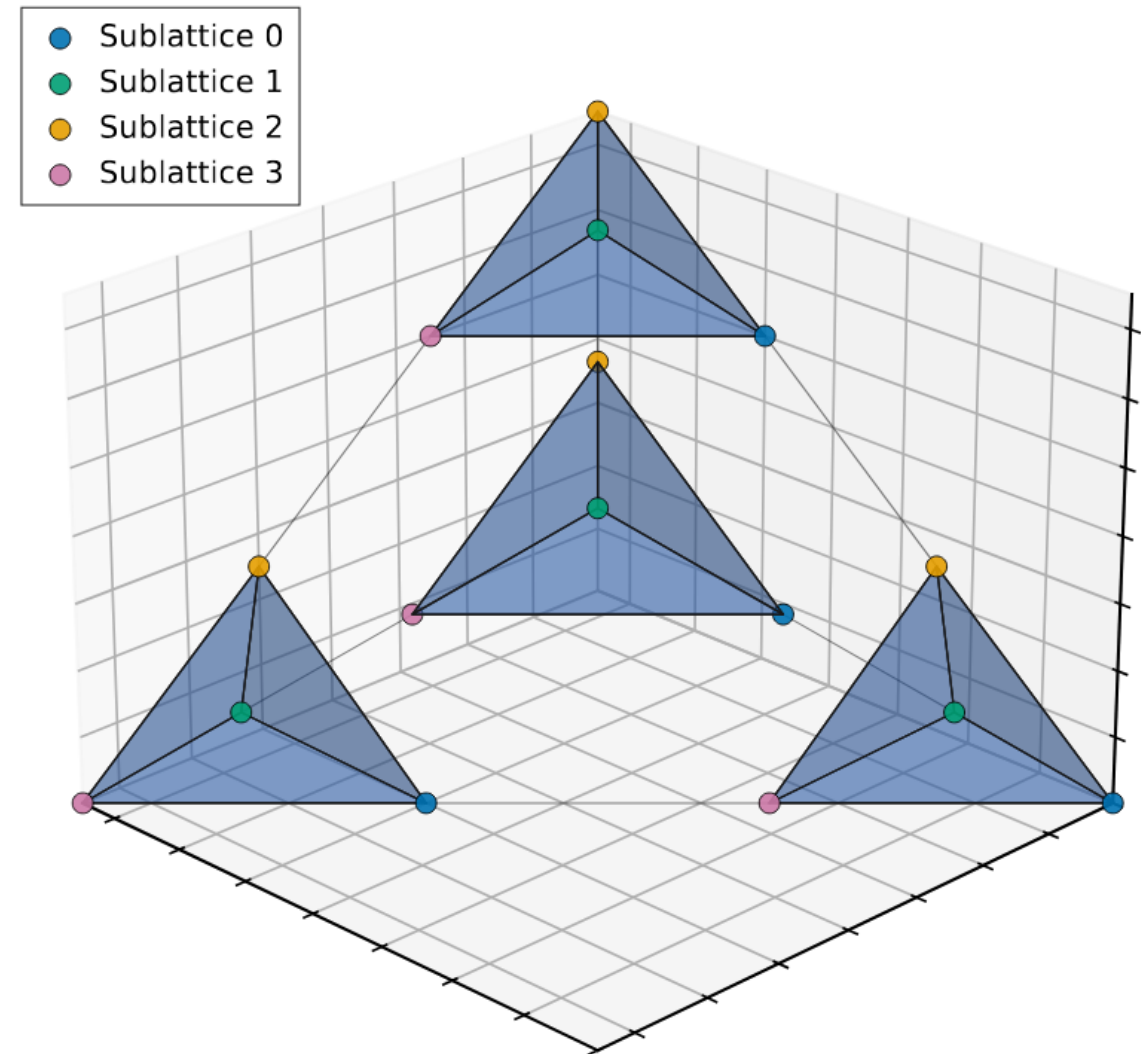
The microcanonical thermal pure quantum method (mTPQ)

Starting from a state $|\psi_0\rangle$ at $T \rightarrow 0$, the real time correlation is measured by

$$\langle \psi_0 | O(t) O(0) | \psi_0 \rangle = \langle \psi_0 | e^{iHt} O e^{-iHt} O | \psi_0 \rangle$$

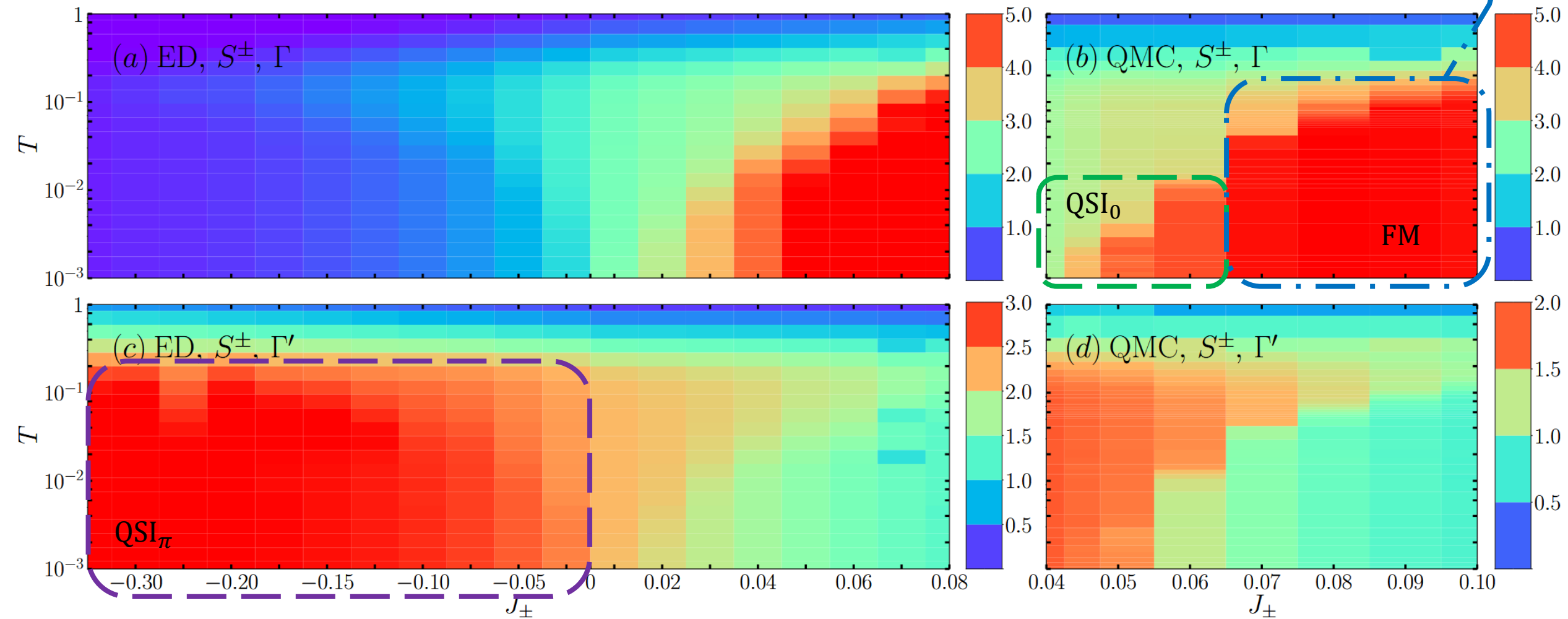
Size limitation: $N = 16$

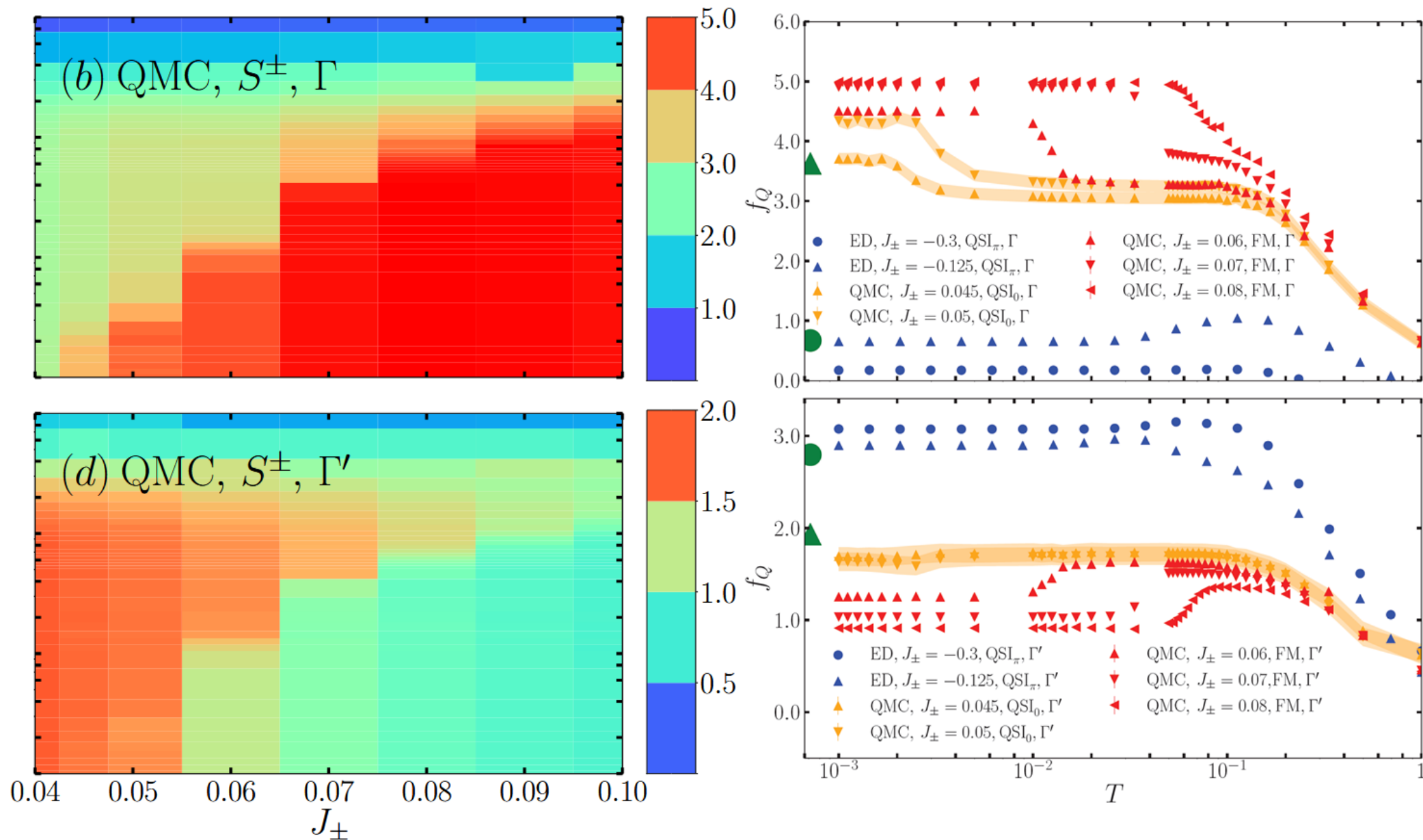
Geometry Preferences



QFI in $S_{\pm}$ channel, $L = 4$
 $\Gamma = (0,0,0)$ and $\Gamma' = (4\pi, 4\pi, 0)$

Understanding: The absence of
spontaneous symmetry breaking
[$U(1)$] in a finite system

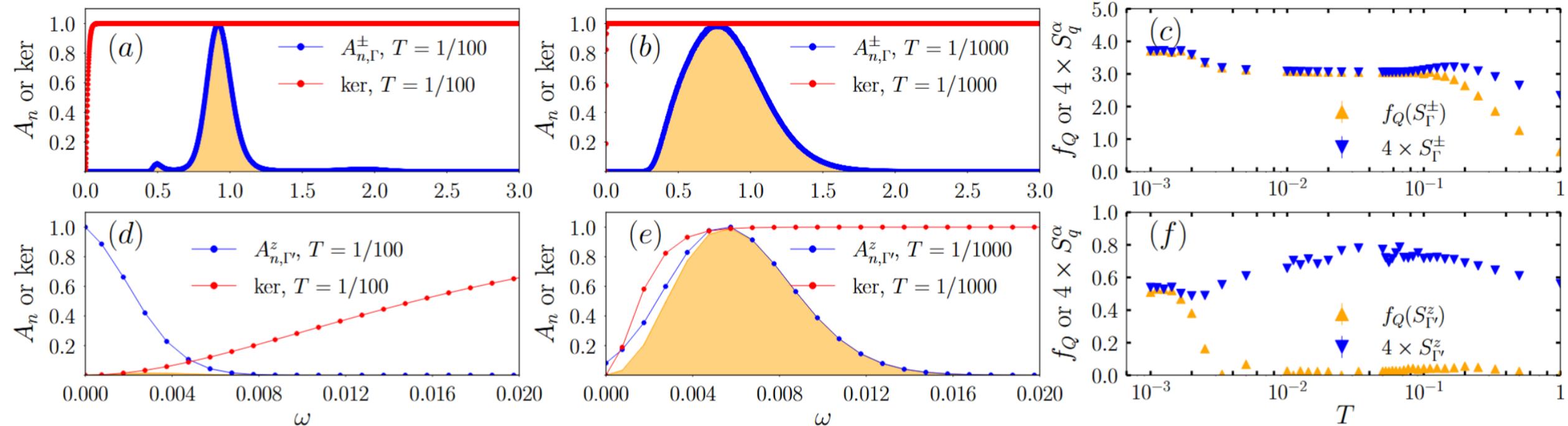




$$\text{Ker} = \tanh\left(\frac{\omega\beta}{2}\right) [1 - \exp(-\beta\omega)]$$

$$f_{QFI} = \int_0^\infty d\omega \, 4\pi \text{Ker} A(\beta, \omega)$$

$$S(\beta) = \int_0^\infty d\omega \, A(\beta, \omega)$$

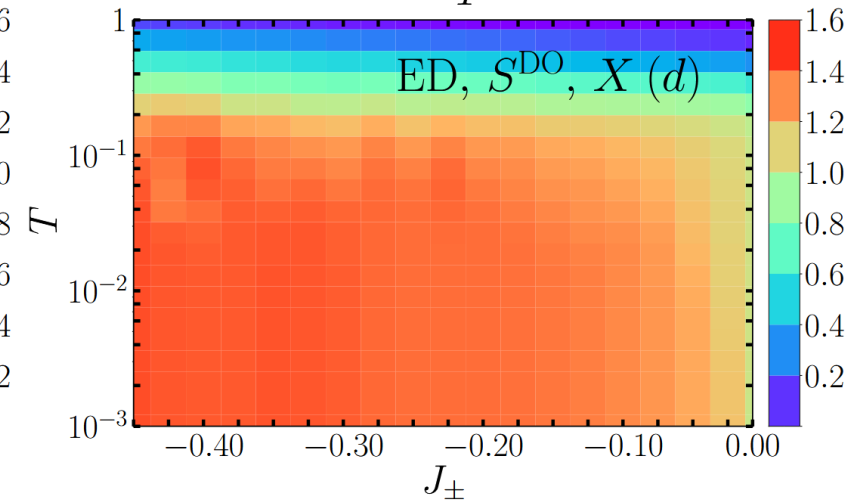
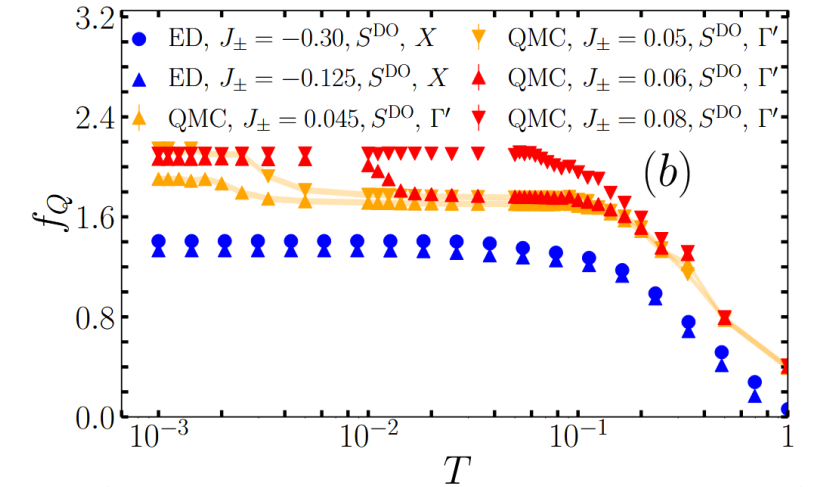
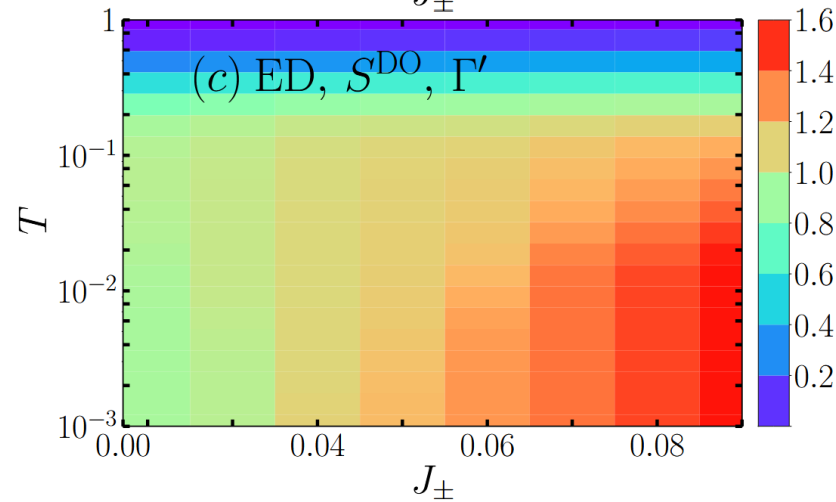
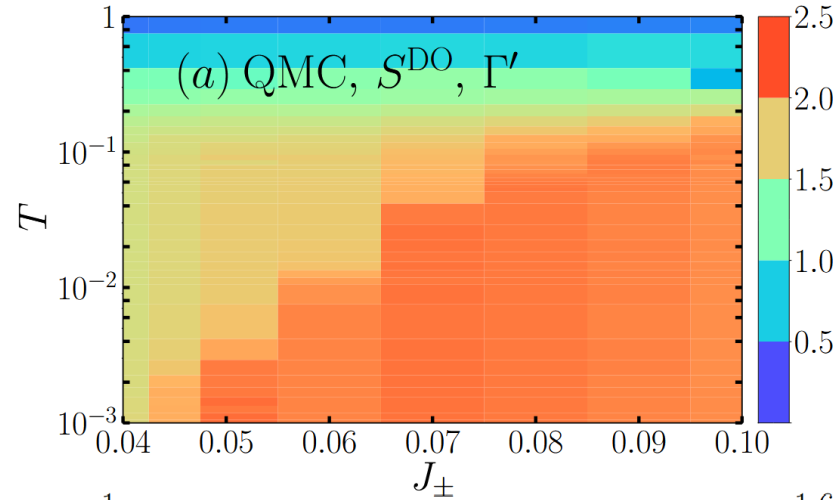
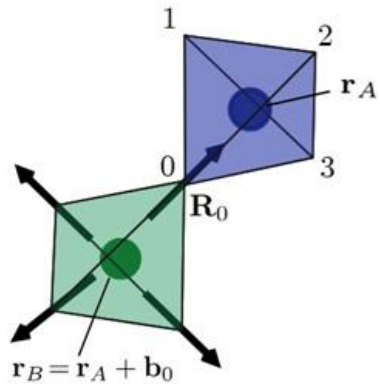


Neutron scattering cross section:
Cerium-based pyrochlore (global frame)

$$A^{DO}(\mathbf{q}, \omega) = \sum_{\mu, \nu} \left(\mathbf{z}_{\mu} \mathbf{z}_{\nu} - \frac{(\mathbf{z}_{\mu} - \mathbf{q})(\mathbf{z}_{\nu} - \mathbf{q})}{|\mathbf{q}|^2} \right) A_{\mu, \nu}^{\pm}(\mathbf{q}, \omega)$$

$$\mathbf{z}_0 = \frac{1}{\sqrt{3}}(1, 1, 1), \mathbf{z}_1 = \frac{1}{\sqrt{3}}(1, -1, -1)$$

$$\mathbf{z}_2 = \frac{1}{\sqrt{3}}(-1, 1, -1), \mathbf{z}_3 = \frac{1}{\sqrt{3}}(-1, -1, 1)$$



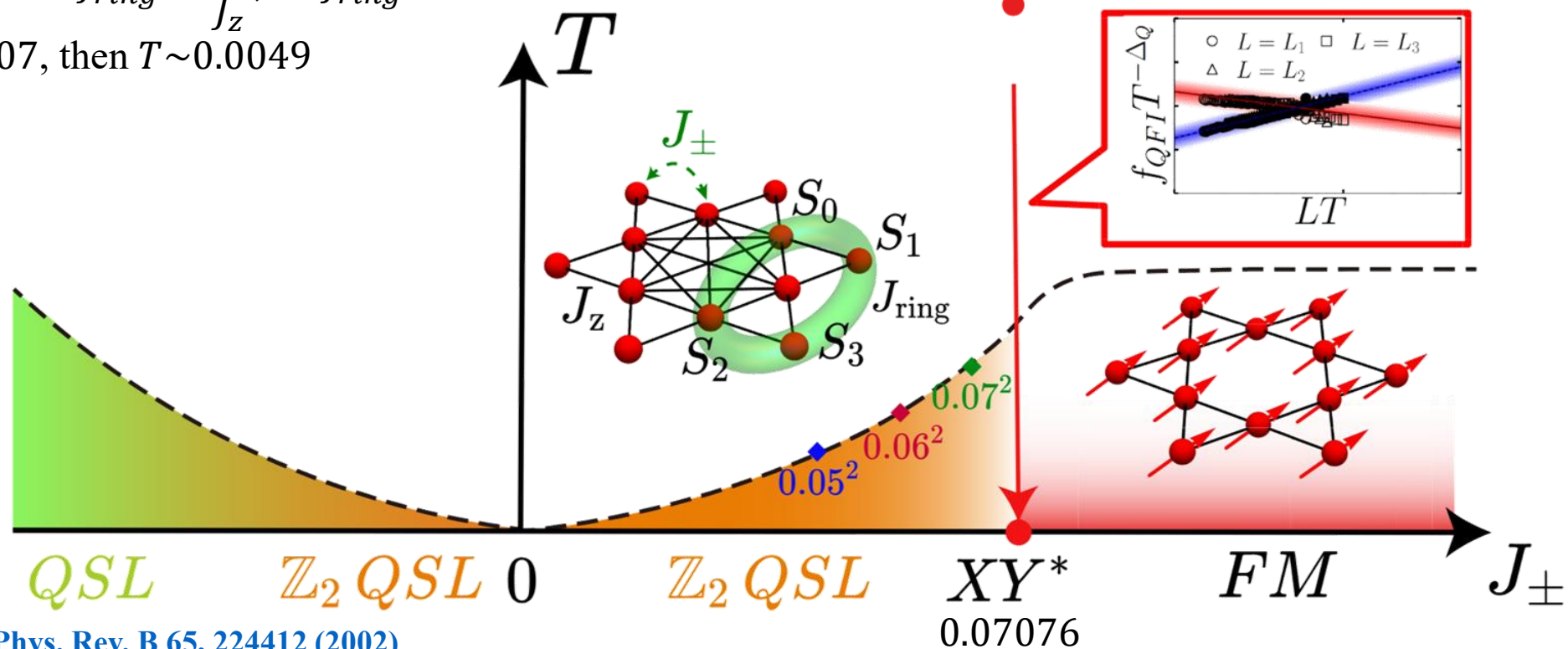
$$H = -J_{\pm} \sum_{\langle i,j \rangle} (S_i^+ S_j^- + S_i^- S_j^+) + \frac{J_z}{2} \sum_{hex} \left(\sum_{i \in hex} S_i^z \right)^2$$

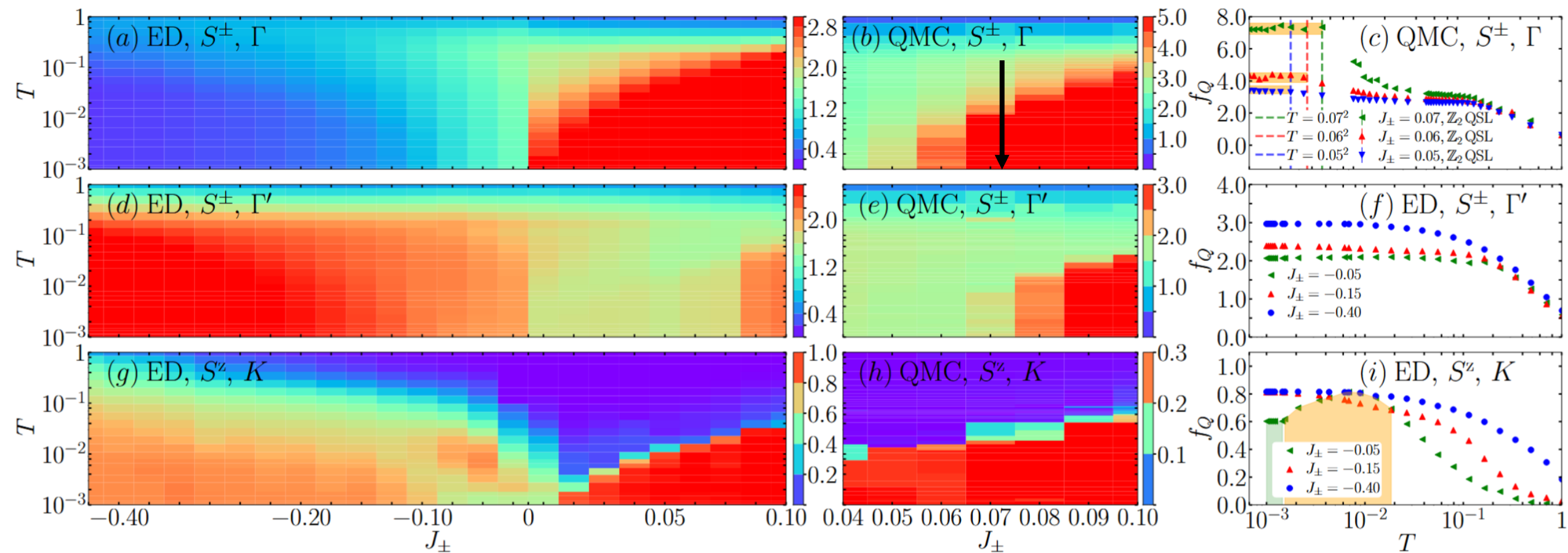
Ring Exchange in BFG model regime:

$$H_{ring} = J_{ring} \sum_{\langle 0,1,2,3 \rangle} (S_0^+ S_1^- S_2^+ S_3^- + S_0^- S_1^+ S_2^- S_3^+)$$

$$J_{ring} = \frac{J_{\pm}^2}{J_z}, T \sim J_{ring}$$

If $J_z = 1, J_{\pm} = 0.07$, then $T \sim 0.0049$





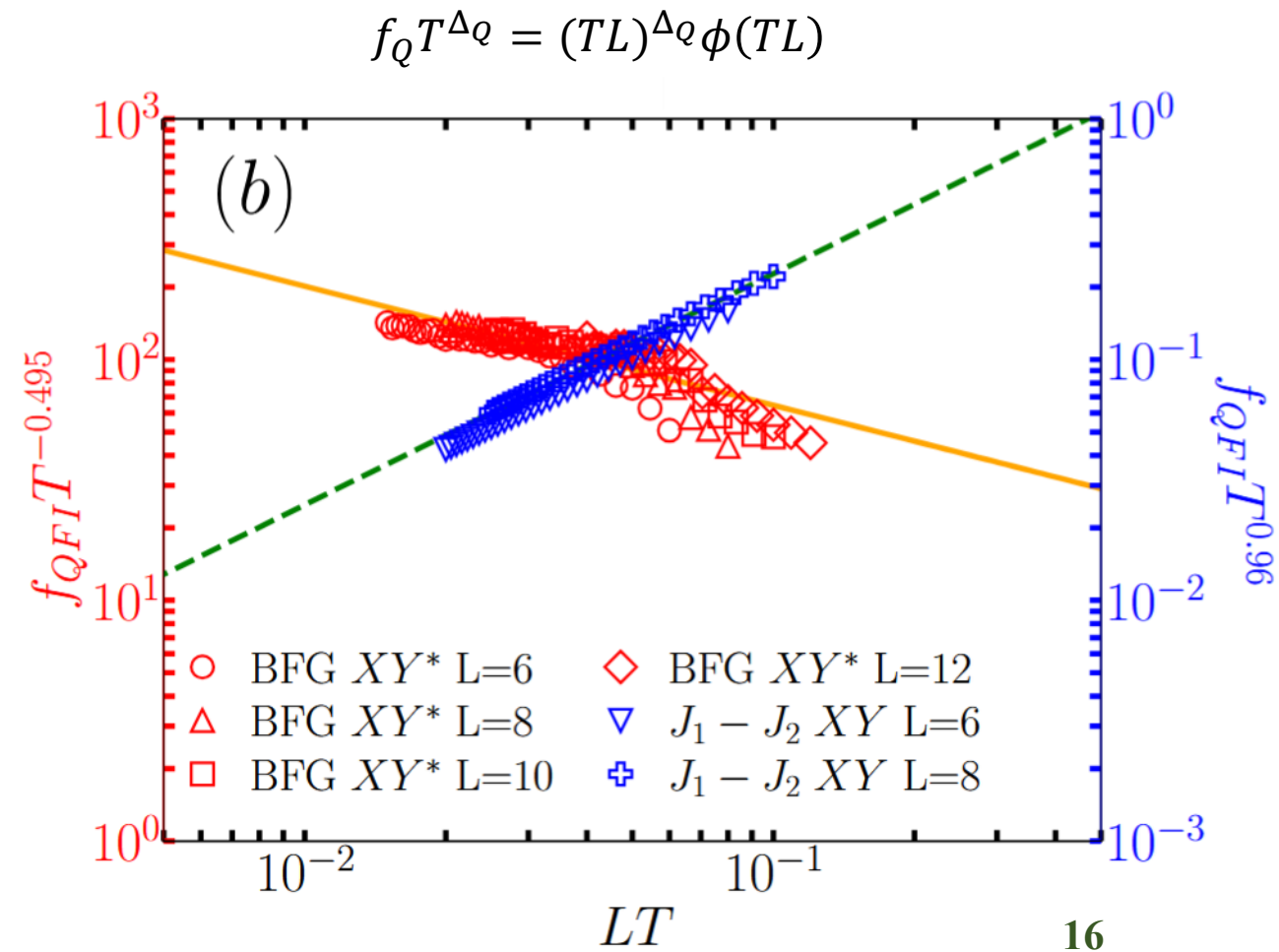
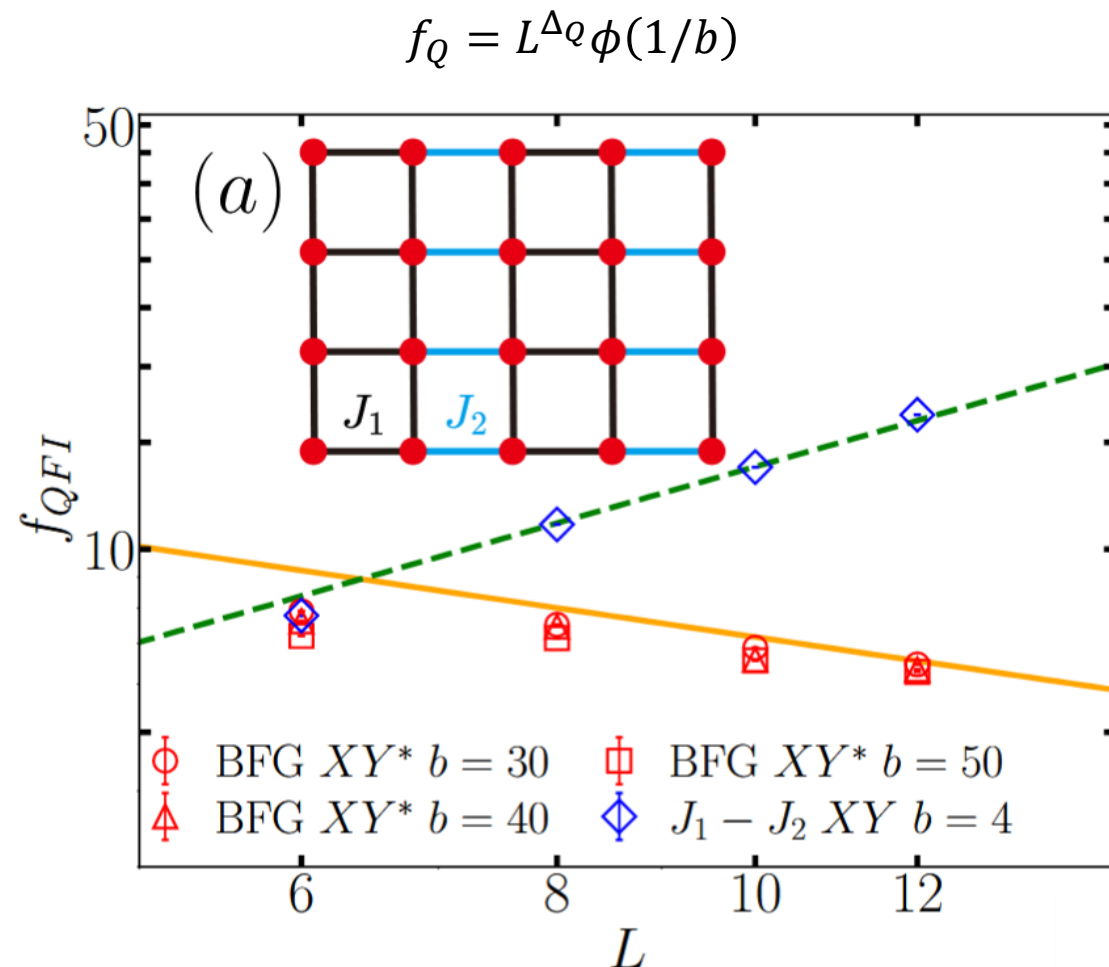
$$f_Q = \lambda^{\Delta_Q} \phi \left(T \lambda^z, L^{-1} \lambda, h \lambda^{\frac{1}{\nu}} \right)$$

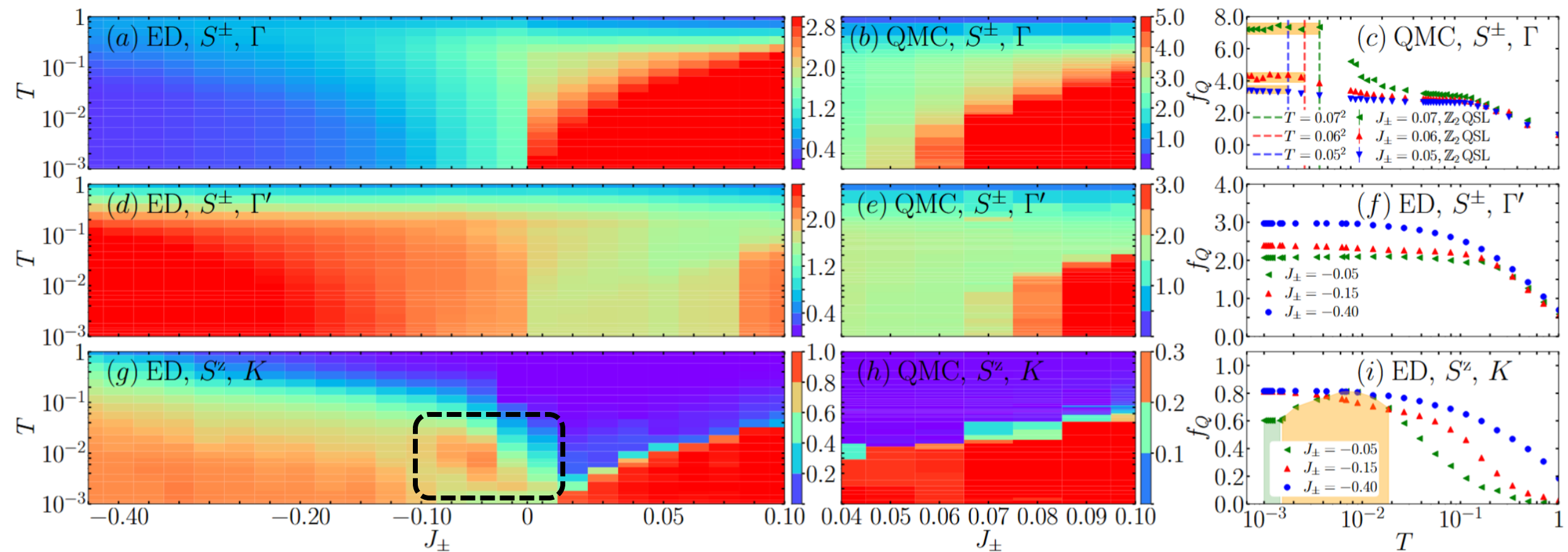
$$h = 0, z = 1, \Delta_Q = 2 - (1 + \eta)$$

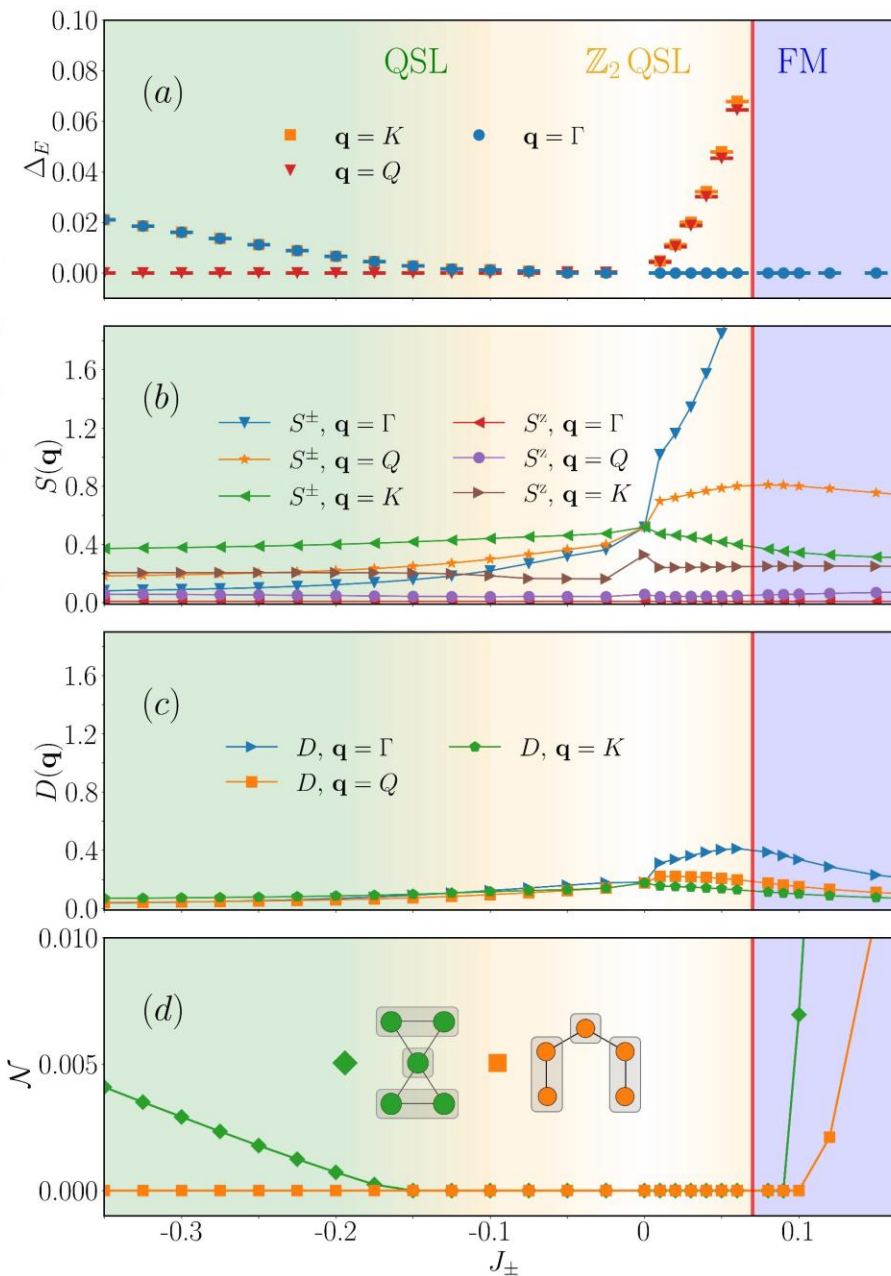
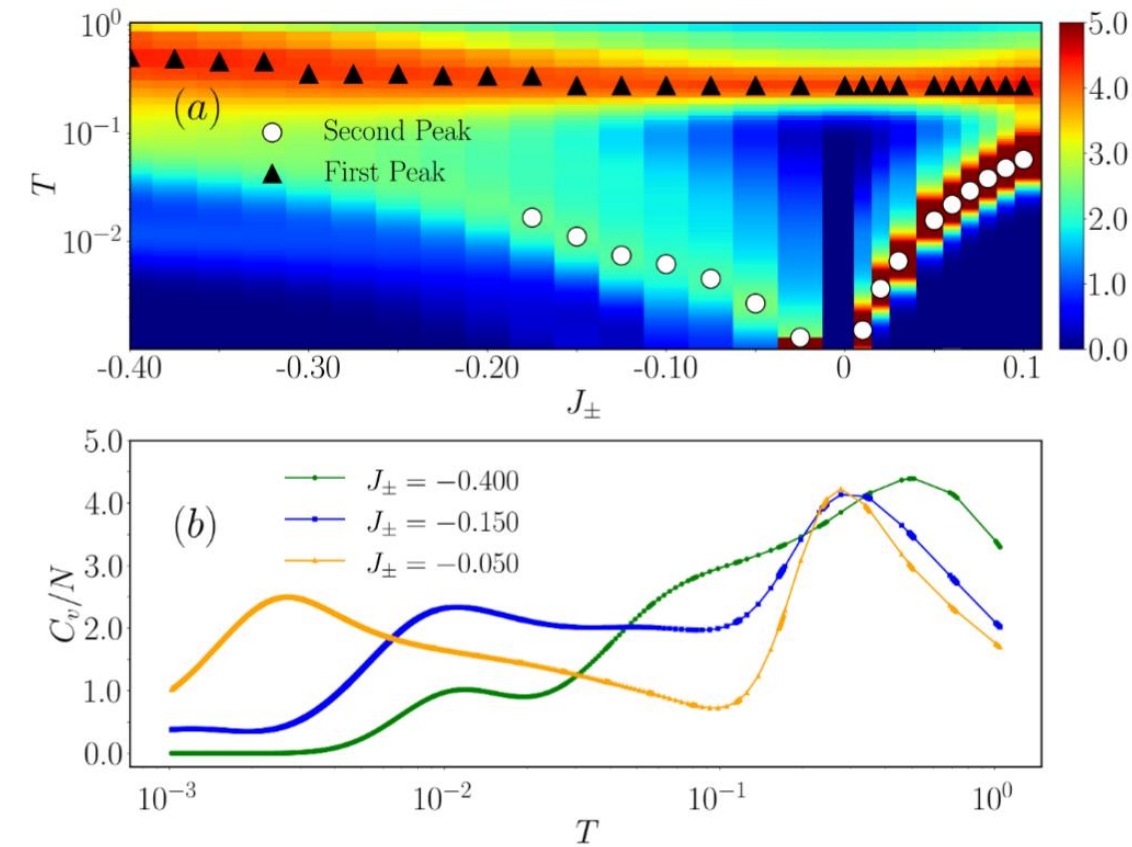
$$\eta \approx 1.495 \text{ (XY*) and } \eta \approx 0.04 \text{ (XY)}$$

$$\Delta_Q = -0.495 \text{ (XY*) and } \Delta_Q = 0.96 \text{ (XY)}$$

$$T \rightarrow 0, \lambda = L, f_Q = \lambda^{\Delta_Q} \phi(T \lambda)$$







Ground state change

No Spin order

No dimer order

Entangle structure change





Thank you.

arXiv:2510.14813 (2025)

arXiv:2603.19951 (2026)



The Quantum Fisher Information is given by

$$F_Q[\rho; \hat{O}(\mathbf{q})] = 2 \sum_{i,j} \frac{(p_i - p_j)^2}{p_i + p_j} |\langle i | \hat{O}(\mathbf{q}) | j \rangle|^2,$$

with $\rho = \sum_i p_i |i\rangle\langle i|$ and $p_i = \exp(-E_i/T)/Z$ referring to a thermal state.

And the dynamic susceptibilities is given by

$$\chi(\omega, \hat{O}(\mathbf{q})) = \sum_{i,j} (p_i - p_j) |\langle i | \hat{O}(\mathbf{q}) | j \rangle|^2 \pi \delta(\omega - E_j + E_i).$$

Noting

$$\int_{-\infty}^{\infty} d\omega \tanh\left(\frac{\omega}{2T}\right) \pi \delta(\omega - E_i + E_j) = \tanh\left(\frac{E_i - E_j}{2T}\right) = \frac{p_j - p_i}{p_j + p_i}.$$

Thus

$$F_Q[\rho; \hat{O}(\mathbf{q})] = \frac{4}{\pi} \int_0^{\infty} d\omega \tanh\left(\frac{\omega}{2T}\right) \chi(\omega, \hat{O}(\mathbf{q})).$$

For given operator $\hat{O}$, such as $\hat{O} = \sum_r e^{iqr} \hat{S}^z(r)$, the corresponding QFI is defined as

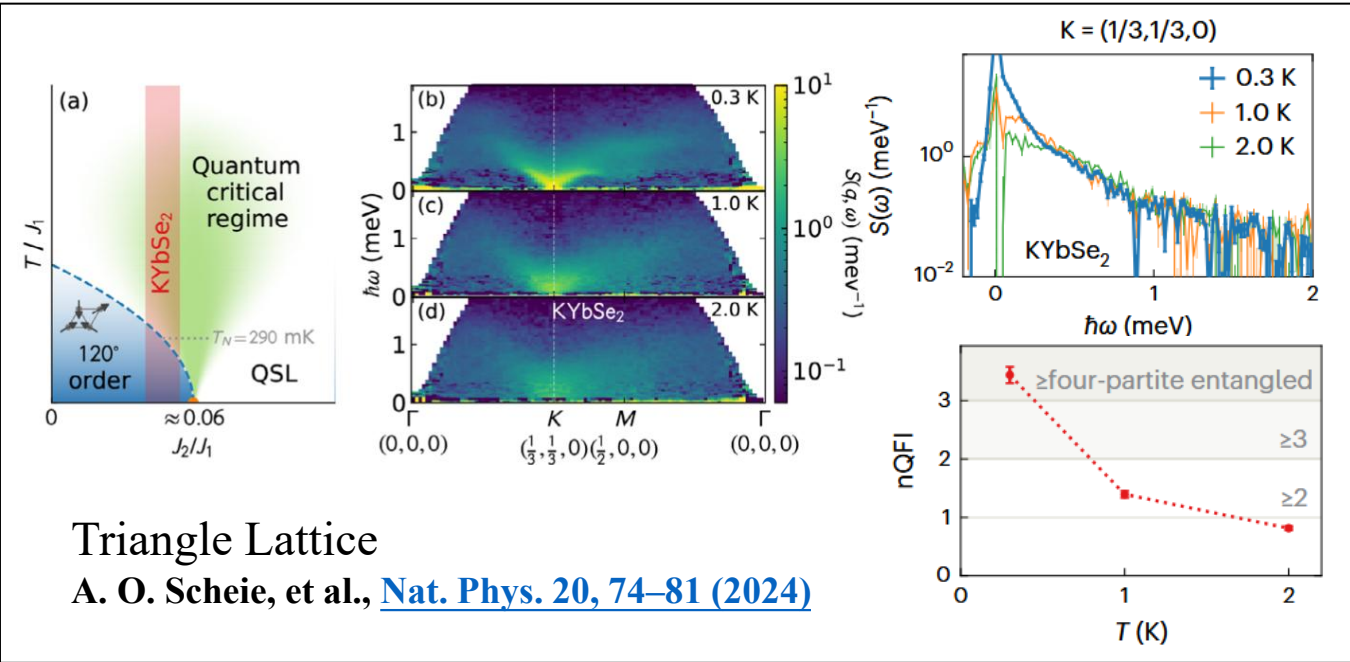
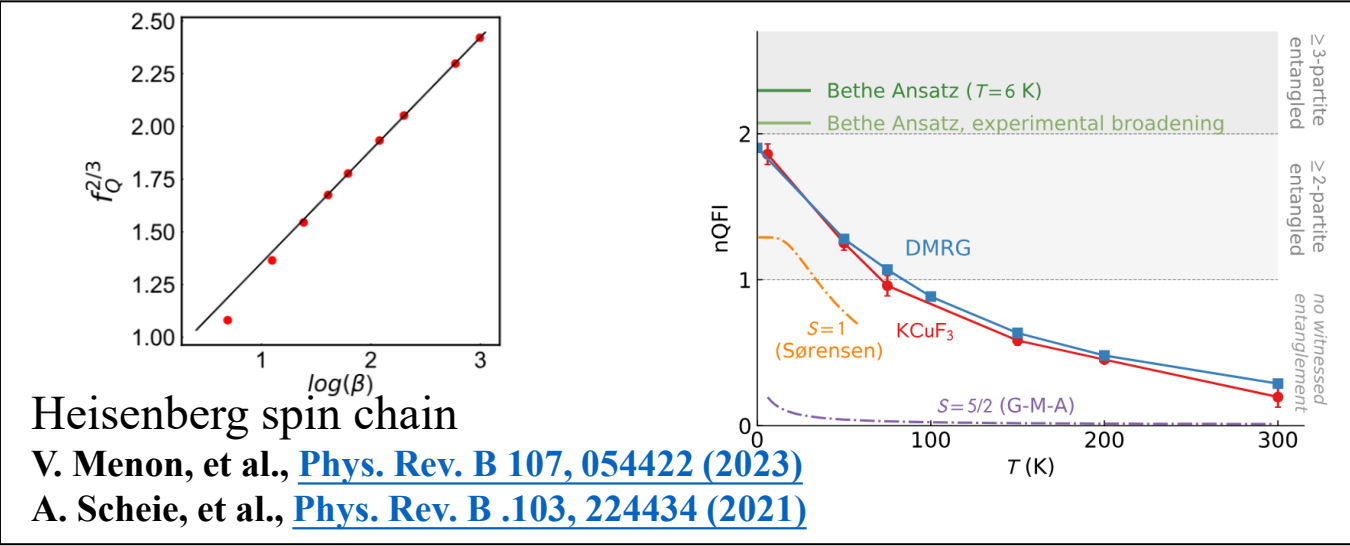
$$f_{QFI} = F_Q/N = \int_0^\infty d\omega \, 4 \tanh\left(\frac{\omega\beta}{2}\right) \pi[1 - \exp(-\beta\omega)] A(\beta, \omega) ,$$

with A is the dynamic structure factor of operator $\hat{O}$,

If $f_{QFI} > m(O_{max} - O_{min})^2$, the corresponding state is at-least $(m + 1)$ -partite entangled.

$$\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) \otimes \frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle), m < 2$$
$$\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle) \otimes |\uparrow\rangle \otimes |\downarrow\rangle, m < 1$$

Philipp Hauke, et al., [Nature Phys 12, 778–782 \(2016\)](#)
Luca Pezzé, et al., [Phys. Rev. Lett. 102, 100401 \(2009\)](#)



What is the physical meaning of Quantum Fisher information's independence of the dynamic spectrum?

Quantum Cramér-Rao bound: Given a state ρ_0 , if a unitary transform is applied as $\rho_\theta = e^{-i\theta G} \rho_0 e^{i\theta G}$, with θ as the parameter to be estimated and G a Hermitian operator, and an observable O is measured to infer θ , then the estimation variance satisfies

$$\text{Var}(\theta) \geq \frac{1}{mF_Q}$$

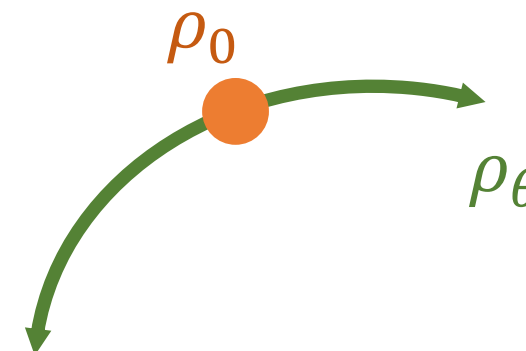
F_Q can be observed from the dynamic spectra of G

m : independent measuring time

F_Q : Quantum Fisher Information:

$$F_Q = N f_{QFI}$$

$$f_{QFI} = \int_0^\infty d\omega \, 4 \tanh\left(\frac{\omega\beta}{2}\right) \pi [1 - \exp(-\beta\omega)] A_G(\beta, \omega)$$



For pure state $|\psi_0\rangle$, operation on two spin, such as $U(\theta) = \exp[-i(\sigma_1^z + \sigma_2^z)\theta]$, the Quantum Fisher Information is given by

$$F_Q = 4 \left[\left\langle \frac{\partial \psi_\theta}{\partial x} \middle| \frac{\partial \psi_\theta}{\partial x} \right\rangle - \left| \left\langle \psi_\theta \middle| \frac{\partial \psi_\theta}{\partial x} \right\rangle \right|^2 \right],$$

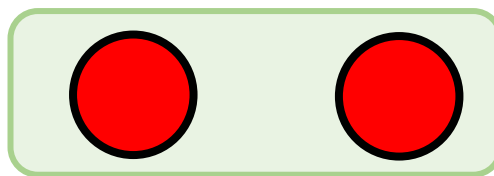
and the Cramér–Rao bound is $\Delta\theta^2 = \text{Var}(\theta) \geq 1/mF_Q$.

Heisenberg limit: Best entangled

$$\frac{1}{\sqrt{2}}(|\uparrow\downarrow\rangle - |\downarrow\uparrow\rangle),$$

$$F_Q = 4$$

$$\Delta\theta \propto \sqrt{\frac{1}{F_Q}} = \frac{1}{N}$$

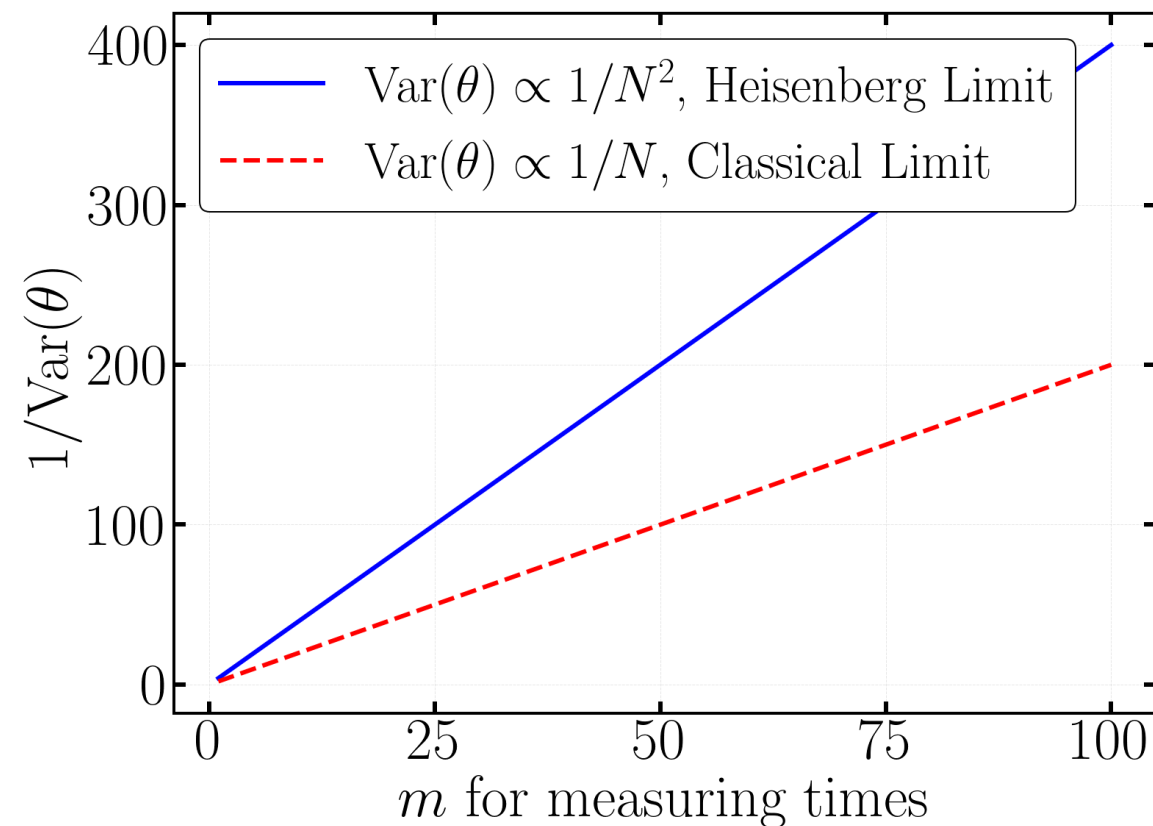


Classical limit: Standard quantum

$$\frac{1}{2}(|\uparrow\rangle + |\downarrow\rangle) \otimes (|\uparrow\rangle + |\downarrow\rangle),$$

$$F_Q = 2$$

$$\Delta\theta \propto \sqrt{\frac{1}{F_Q}} = \frac{1}{\sqrt{N}}$$



For $A^z(\mathbf{q}, \omega)$, $(O_{max} - O_{min})^2 = \left(\frac{1}{2} - \left(-\frac{1}{2}\right)\right)^2 = 1$, therefore

$$f_{QFI, A^z} < m$$

For $A^\pm(\mathbf{q}, \omega) = A^x(\mathbf{q}, \omega) + A^y(\mathbf{q}, \omega)$, $(O_{max} - O_{min})^2 = 2 \left(\frac{1}{2} - \left(-\frac{1}{2}\right)\right)^2 = 2$, therefore

$$f_{QFI, A^\pm} < \frac{m}{2}$$

For $A^{DO}(\mathbf{q}, \omega) = A^{NSF}(\mathbf{q}, \omega) + A^{SF}(\mathbf{q}, \omega)$, where (SF/NSF) means spin-flip/non-spin-flip channel with respect to the neutron polarization vector $\mathbf{p}$. For $A^{NSF/SF}(\mathbf{q}, \omega)$,

$$(O_{max} - O_{min})^2 = \frac{1}{4} \sum_{\mu} \left| \Pi_{(N)SF}^{\mu} \right|^2,$$

where

$$\Pi_{NSF}^{\mu} = \mathbf{p} \mathbf{z}_{\mu}, \quad \Pi_{SF}^{\mu} = \mathbf{v} \mathbf{z}_{\mu}, \quad \mathbf{p} \mathbf{v} = 0$$

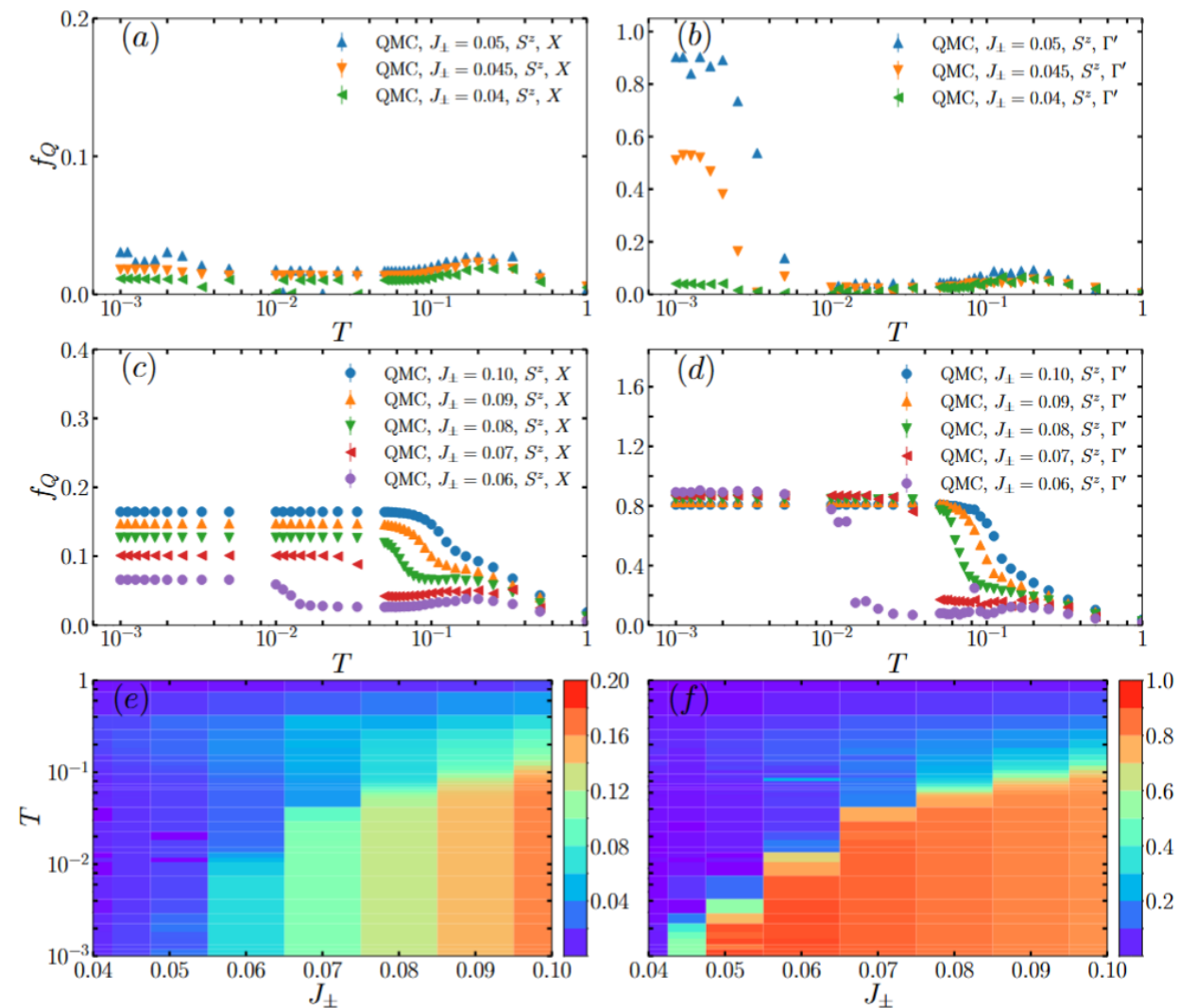
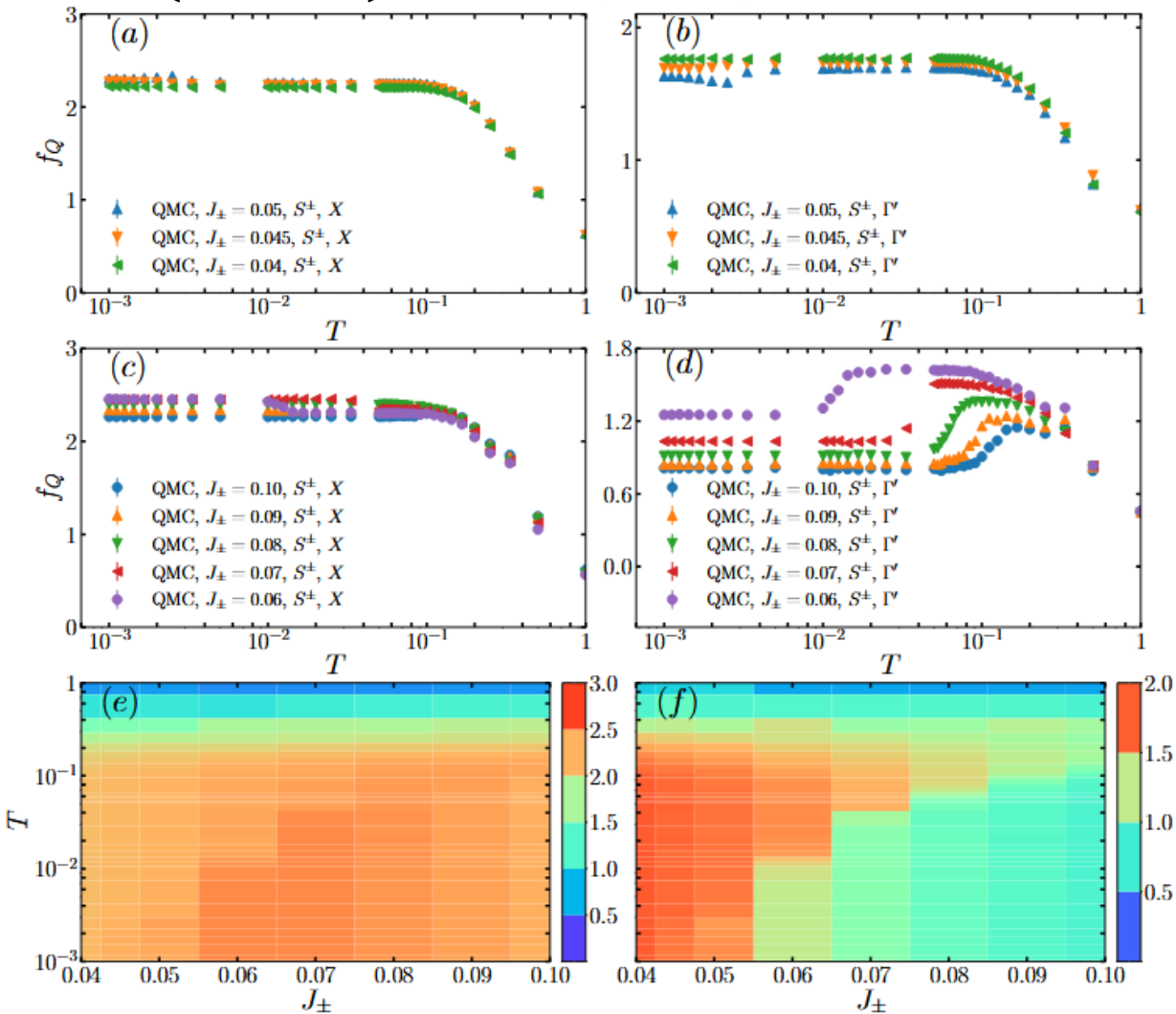
Thus

$$f_{QFI, A^{(N)SF}} < m \text{Tr} \left[\sum_{\mu} \mathbf{p}^T \mathbf{z}_{\mu}^T \mathbf{z}_{\mu} \mathbf{p} \right] = \frac{m}{3}$$

$$f_{QFI, A^{DO}} < \frac{2m}{3}$$

$L = 4$

$\Gamma' = (4\pi, 4\pi, 0)$ and $X = (0, 0, 2\pi)$



QFI in $S_{\pm}$ channel, $L = 4$
 $\Gamma = (0,0,0)$ and $\Gamma' = (4\pi, 4\pi, 0)$

Understanding (1): The absence of spontaneous symmetry breaking
 $[U(1)]$ in a finite system
 Understanding (2): Operator chosen

